

Theoretical pbs - Exercise (1)

$$\boxed{1.} \quad \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$$

$$\lambda = |CA| : |CB|$$

$$\frac{CA}{CB} = \lambda \Rightarrow AC = \lambda \cdot x; CB = x$$

$$A, C, B \text{ collinear} \Leftrightarrow AC = \lambda \cdot CB$$

$$\Rightarrow \vec{OC} - \vec{OA} = \lambda (\vec{OB} - \vec{OC})$$

$$\Rightarrow \vec{OC} + \lambda \vec{OC} = \vec{OA} + \lambda \vec{OB}$$

$$\Rightarrow (1+\lambda) \vec{OC} = \vec{OA} + \lambda \vec{OB} \quad | : (1+\lambda)$$

$$\Rightarrow \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$$

$$\Rightarrow C = \frac{1}{1+\lambda} (A + \lambda B)$$

$$\Rightarrow \begin{cases} c_1 = \frac{a_1 + \lambda \cdot b_1}{\lambda + 1} \end{cases}$$

$$\begin{cases} c_2 = \frac{a_2 + \lambda \cdot b_2}{\lambda + 1} \end{cases}$$

$\vdots$

$$\begin{cases} c_m = \frac{a_m + \lambda \cdot b_m}{\lambda + 1} \end{cases}$$

$\boxed{2.}$ Tetrahedron ABCD; How many frames?

origin $\rightarrow 4$ ways

$$K = (\text{origin}, v_1, v_2, v_3) \quad \mathbb{R}^3$$

$\downarrow$
4

$\downarrow$
3! ways

$$\Rightarrow 4 \cdot 6 = 24$$

①

[3]. K, K' .

$$\begin{aligned} [P]_{K'} &= M_{B'B} \cdot ([P]_K - [O']_K) \\ &= M_{B'B}^{-1} ([P]_K - [O']_K) \\ &= M_{B'B} \cdot [P]_K + [O]_{K'} \end{aligned}$$

$$\bullet [P]_{K'} = [O'P]_{B'} = [OP]_B - [OO']_B = [P]_K - [O']_K.$$

$$[P]_K = [OP]_B$$

$$[O']_K = [OO']_B.$$

$$\begin{aligned} \bullet [P]_{K'} &= [O'P]_{B'} = M_{B'B} \cdot [OP]_B \\ &= M_{B'B} \cdot ([P]_K - [O']_K) \\ &= M_{B'B} \cdot [P]_K - M_{B'B} \cdot [O']_K \\ &= M_{B'B} [P]_K - M_{B'B} \cdot [OO']_B \\ &= M_{B'B} \cdot [P]_K - [OO']_{B'} \\ &= M_{B'B} \cdot [P]_K + [O'O]_{B'} \\ &= \dots + [O]_{K'} \end{aligned}$$

[4]. $l_1: \begin{cases} a_1x + a_2y + \dots \\ b_1x + b_2y + \dots \end{cases}$

$l_2: \begin{cases} c_1x + c_2y + \dots \\ d_1x + d_2y + \dots \end{cases}$

l_1, l_2 coplanare $\Leftrightarrow \begin{vmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{vmatrix} = 0$

$$\bar{u}_1: a_1 x + \dots = 0 \quad \Rightarrow l_1 = \pi_1 \cap \bar{u}_2$$

$$\overline{u}_2: b_1 x + \dots = 0$$

$$\bar{u}_3: 01x + \dots = 0 \quad ? \Rightarrow l_2 = \bar{u}_3 \cap \bar{u}_4.$$

$$\bar{u}_2: d_1 x + \dots = 0$$

$\Rightarrow$ l_1, l_2 coplanar (hyp.)

$$\det(\tilde{\pi}) = 0 \text{ (cl.)}$$

- case 1: $l_1 \cap l_2 \neq \emptyset$.

$$\Rightarrow \begin{cases} l_1 \\ l_2 \end{cases} \text{ has unique sol.}$$

$$\Rightarrow \text{rang } M = \text{rang } \tilde{M} = \dots \Rightarrow \text{rang } M \leq 3. \quad \textcircled{\neq 4}$$

$$\Rightarrow \det(\tilde{K}) = 0.$$

- Case 2: $h_1 \parallel h_2$

Def.
($\Rightarrow$) $D(l_1) = D(l_2) \Leftrightarrow \langle v_1 \rangle = \langle v_2 \rangle.$

2) $D(l_1), D(l_2)$ solutions.

$$\Rightarrow \tilde{M} \sim \text{echelon} \Rightarrow \text{rank } \tilde{M} < 4 \Rightarrow \det \tilde{M} = 0.$$

$$\iff \det(\tilde{M}) = 0 \text{ (hyp.)}$$

h, l2 coplemar (cl.)

$$\det(\tilde{M}) = 0 \Rightarrow \text{rank}(\tilde{M}) \leq 3.$$

$$l_1, l_2 \text{ lines} \Rightarrow \text{rank} \geq 2.$$

$$\bullet \text{ case 1: } \text{rank } M = \text{rank } \tilde{M} = 3.$$

$$\Rightarrow \cap^{\text{in}} \text{ one point.}$$

$$\Rightarrow \begin{cases} l_1 \\ l_2 \end{cases} \text{ has unique sol.}$$

$$\Rightarrow \exists P \in l_1 \cap l_2 \Rightarrow l_1, l_2 \text{ coplanar. } (\cap).$$

$$\bullet \text{ case 2: } \text{rank}(M) = 2; \text{rank}(\tilde{M}) = 3.$$

$$\Rightarrow D(l_1) = D(l_2) \Rightarrow l_1 \parallel l_2.$$

$$\Rightarrow \text{coplanar.}$$

$$\boxed{5} \quad \begin{cases} \bar{u}_1: a_1 x \dots \\ \bar{u}_2: b_1 x \dots \\ \bar{u}_3: c_1 x \dots \end{cases}$$

By theorem 2.6. $\Rightarrow$ if (5) compatible,
the sol. set. is an affine subspace of A^3
with $\dim 3 - \text{rank}(M)$.

$$(5) \text{ comp} \Leftrightarrow \text{rank}(M) = \text{rank}(\tilde{M})$$

$$(5) \text{ incomp} \Rightarrow \emptyset.$$

case 1: $\text{rank } K \subset \text{rank } \tilde{M}$

$$\Rightarrow \emptyset.$$

case 2: $\text{rank } K = \text{rank } \tilde{M} = 3$

$$\Rightarrow (S) \text{ comp.} \Rightarrow \text{sol. has dim } 3-3=0.$$

$$\Rightarrow \bar{u}_1 \cap \bar{u}_2 \cap \bar{u}_3 = \{P\}$$

$\Downarrow$
point.

$$P(x, y, z).$$

case 3: $\text{rank } K = \text{rank } \tilde{M} = 2$

$$(S) \text{ comp.} \Rightarrow \text{dim } 3-2=1. \Rightarrow \text{line.}$$

$$\Rightarrow \bar{u}_1 \cap \bar{u}_2 \cap \bar{u}_3 = \{l\}$$

$$l: ax + by + cz = 0.$$

case 4: $\text{rank } K = \text{rank } \tilde{M} = 1$

$$(S) \text{ comp.} \Rightarrow \text{dim } 3-1=2.$$

$$\Rightarrow \bar{u}_1 \cap \bar{u}_2 \cap \bar{u}_3 = \{\bar{u}_4\}$$

$$\bar{u}_4: ax + by + cz + d = 0.$$

6. Cartesian $\rightleftharpoons$ parametric

plane in cartesian: $\bar{u}: ax + by + cz + d = 0$

plane in parametric: $(x, y, z) = P_0 + s \cdot \bar{u} + t \cdot \bar{v}$

$(\mathbb{A}^3)$

• $C \Rightarrow P$

• $P_0 : y=0, z=0.$

$$\Rightarrow ax+d=0 \Rightarrow x = -\frac{d}{a} \quad (a \neq 0).$$

$$\Rightarrow P_0(x_0, y_0, z_0) = P\left(-\frac{d}{a}, 0, 0\right).$$

• dir. vector: $ax+by+cz=0.$

$$y=1, z=0 \Rightarrow ax+b=0 \Rightarrow x = -\frac{b}{a}$$

$$\Rightarrow \vec{u} \left(-\frac{b}{a}, 1, 0\right).$$

$$y=0, z=1 \Rightarrow ax+c=0 \Rightarrow x = -\frac{c}{a}$$

$$\Rightarrow \vec{v} \left(-\frac{c}{a}, 0, 1\right).$$

• $\vec{u}, \vec{v}$ lin. ind!

$$\alpha \vec{v} + \beta \vec{u} = 0$$

$$\Rightarrow \alpha \left(-\frac{c}{a}, 0, 1\right) + \beta \left(-\frac{b}{a}, 1, 0\right) = 0.$$

$$\Rightarrow \left(-\alpha \cdot \frac{c}{a} - \beta \cdot \frac{b}{a}, \beta, \alpha\right) = 0$$

$$\Rightarrow \begin{cases} -\alpha \cdot \frac{c}{a} - \beta \cdot \frac{b}{a} = 0 \\ -\beta \cdot \frac{b}{a} = 0 \\ \alpha = 0 \end{cases} \quad \underline{\underline{\beta = 0}}$$

$\Rightarrow \vec{u}, \vec{v}$ lin. ind.

$$\Rightarrow (x, y, z) = P_0 + s \cdot \vec{v} + t \cdot \vec{u}$$

$$= \left(-\frac{d}{a}, 0, 0\right) + s \cdot \left(-\frac{b}{a}, 1, 0\right)$$

$$+ t \cdot \left(-\frac{c}{a}, 0, 1\right).$$

$\varphi \rightarrow c$

$$\vec{n} = \vec{u} \times \vec{v}, \quad \vec{n}(a, b, c)$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$ax + by + cz - \underbrace{(ax_0 + by_0 + cz_0)}_d = 0$$

$$\Rightarrow ax + by + cz + d = 0$$

2.6. K in A^m

$S =$ affine space of dimension $m-k$.

$$(S): \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tn}x_n = b_t \end{cases}$$

$$D(S): \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tn}x_n = 0 \end{cases}$$

$D(S)$ vector subspace.

$u \Rightarrow S = \{x_1, \dots, x_m\}$ (hyp.)

$S \rightarrow$ affine subspace of A^m (cl.)

$$S = U + P$$

$P \rightarrow$ solution from S ; $P(p_1, \dots, p_m)$.

$U \rightarrow$ vector subspace.

$S = T$; $T \rightarrow$ solution from S .

$$\begin{cases} S \subseteq T \\ T \subseteq S \end{cases}$$

$Q \rightarrow$ solution from S .

$$Q(g_1, \dots, g_m) \in S.$$

$$Q \in T \Leftrightarrow \vec{PQ} \in U.$$

$$a_{j1}(g_1 - p_1) + \dots + a_{jm}(g_m - p_m) = 0.$$

$$\underbrace{a_{j1}g_1 + \dots + a_{jm}g_m}_{b_j} - \underbrace{(a_{j1}p_1 + \dots + a_{jm}p_m)}_{b_j} = 0.$$

$$\Rightarrow \vec{PQ} \in U.$$

$$\Rightarrow S \subseteq T. \quad | \quad \Rightarrow S = T.$$

$$T \subseteq S.$$

" $\Leftarrow$ " Define $f: \mathbb{R}^m \rightarrow \mathbb{R}$,

$$f(x_1, \dots, x_m) = x_1 w_1 + \dots + x_d w_d. \quad (\text{lin. map.})$$

$$\dim \mathbb{R}^m = \dim \ker f + \dim \text{Im} f.$$

$$\text{Im} f = \langle w_1, \dots, w_d \rangle$$

$$\ker f = \{ (x_1, \dots, x_m) \in \mathbb{R}^m \mid f(x_1, \dots, x_m) = 0 \}$$

$$= \{ (x_1, \dots, x_m) \in \mathbb{R}^m \mid \langle x_1 v_1 + \dots + x_m v_m \rangle = 0 \}.$$

$$v_1 = \begin{pmatrix} v_{11} \\ \vdots \\ v_{m1} \end{pmatrix}, \quad v_m = \begin{pmatrix} v_{1m} \\ \vdots \\ v_{mm} \end{pmatrix}$$

$$\Rightarrow \begin{cases} x_1 \cdot \underline{v_{11}} + \dots + x_m \cdot \underline{v_{m1}} = 0 \\ \vdots \\ x_1 \cdot \underline{v_{1m}} + \dots + x_m \cdot \underline{v_{mm}} = 0. \end{cases}$$

$$\dim \ker f = m - d.$$

2) keep $m - d$ eq.

3) rank... = $m - d$.

Q → solution from S.

$$Q(g_1, \dots, g_m) \in S.$$

$$Q \in T \Leftrightarrow \vec{PQ} \in U.$$

$$a_{j1}(g_1 - p_1) + \dots + a_{jm}(g_m - p_m) = 0.$$

$$= \underbrace{a_{j1}g_1 + \dots + a_{jm}g_m}_{b_j} - \underbrace{(a_{j1}p_1 + \dots + a_{jm}p_m)}_{b_j} = 0.$$

$$\Rightarrow \vec{PQ} \in U.$$

$$\Rightarrow S \subseteq T. \quad | \quad \Rightarrow S = T.$$

$$T \subseteq S.$$

u $\hookrightarrow$ Define $f: \mathbb{R}^m \rightarrow \mathbb{R}^d$,

$$f(x_1, \dots, x_m) = x_1 w_1 + \dots + x_d w_d. \quad (\text{lin. map})$$

$$\dim \mathbb{R}^m = \dim \ker f + \dim \text{Im} f.$$

$$\text{Im} f = \langle w_1, \dots, w_d \rangle$$

$$\ker f = \{ (x_1, \dots, x_m) \in \mathbb{R}^m \mid f(x_1, \dots, x_m) = 0 \}$$

$$= \{ (x_1, \dots, x_m) \in \mathbb{R}^m \mid \langle x_1 v_1 + \dots + x_m v_m \rangle = 0 \}.$$

$$v_1 = \begin{pmatrix} v_{11} \\ \vdots \\ v_{m1} \end{pmatrix}, \quad v_m = \begin{pmatrix} v_{1m} \\ \vdots \\ v_{mm} \end{pmatrix}.$$

$$\Rightarrow \begin{cases} x_1 \cdot \underline{v_{11}} + \dots + x_m \cdot \underline{v_{m1}} = 0 \\ \vdots \\ x_1 \cdot \underline{v_{1m}} + \dots + x_m \cdot \underline{v_{mm}} = 0. \end{cases}$$

$$\dim \ker f = m - d.$$

2) keep $m - d$ eq.

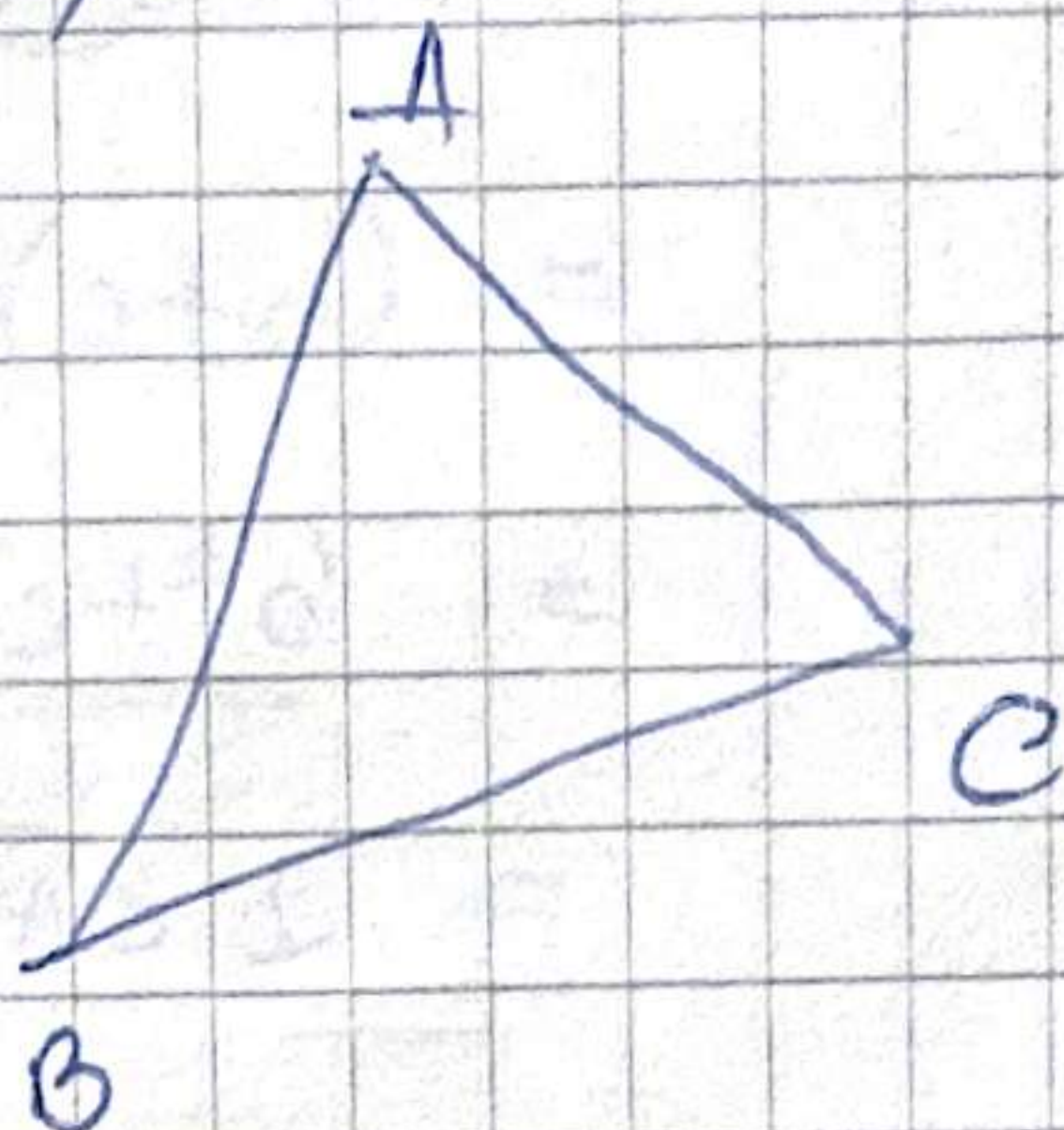
2) rank... = $m - d$.

$$\Delta \text{? } \langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2} (b^2 + c^2 - a^2).$$

? cos law

$$\vec{BC} = \vec{AC} - \vec{AB}.$$

$$|BC| = a$$



$$\Rightarrow |BC|^2 = a^2 = \langle \vec{BC}, \vec{BC} \rangle = \langle (\vec{AC} - \vec{AB}), (\vec{AC} - \vec{AB}) \rangle$$

$$= \langle \vec{AC}, \vec{AC} \rangle - \langle \vec{AC}, \vec{AB} \rangle - \langle \vec{AB}, \vec{AC} \rangle + \langle \vec{AB}, \vec{AB} \rangle$$

$$= |\vec{AC}|^2 + |\vec{AB}|^2 - 2 \langle \vec{AB}, \vec{AC} \rangle$$

$$= b^2 + c^2 - 2 \langle \vec{AB}, \vec{AC} \rangle$$

$$\Rightarrow \langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2} (b^2 + c^2 - a^2)$$

$$\langle \vec{AB}, \vec{AC} \rangle = |\vec{AB}| \cdot |\vec{AC}| \cdot \cos \alpha = \frac{1}{2} (b^2 + c^2 - a^2)$$

$$\Rightarrow c \cdot b \cdot \cos \alpha = \frac{1}{2} (b^2 + c^2 - a^2) \quad | \cdot 2$$

$$\Rightarrow 2bc \cos \alpha = b^2 + c^2 - a^2$$

$$\Rightarrow \underline{a^2 = b^2 + c^2 - 2bc \cos \alpha.}$$

$$\boxed{9} \quad \langle \vec{AB}, \vec{AC} \rangle + bc = 2p(p-a)$$

$$\langle \vec{AB}, \vec{AC} \rangle - bc = -2(p-b)(p-c).$$

$$p = \frac{1}{2} (a+b+c)$$

$$p-a = \frac{1}{2} a + \frac{1}{2} b + \frac{1}{2} c - a = \frac{1}{2} (-a+b+c) \quad | \cdot 4p$$

$$\Rightarrow 4p(p-a) = 2p(-a+b+c)$$

$$= 2 \cdot \frac{1}{2} (a+b+c)(-a+b+c)$$

$$4p(p-a) = (a+b+c)(-a+b+c)$$

$$= (b+c)^2 - a^2$$

$$= \underline{b^2 + c^2 + 2bc} - \underline{a^2}$$

$$= 2 \langle \vec{AB}, \vec{AC} \rangle + 2bc \quad | :2$$

$$\langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2} (b^2 + c^2 - a^2)$$

$$\Rightarrow 2p(p-a) = \langle \vec{AB}, \vec{AC} \rangle + bc.$$

$$p-b = \frac{1}{2} (a-b+c) \quad | \cdot 4(p-c)$$

$$4(p-b)(p-c) = \cancel{2} (a-b+c)(a+b-\underline{c})$$

$$= \cancel{2} (a^2 - (b-c)^2)$$

$$= \cancel{2} (a^2 - b^2 - c^2 + 2bc)$$

$$= \cancel{2} (-2 \langle \vec{AC}, \vec{AB} \rangle + 2bc) \quad | :2$$

$$2(p-b)(p-c) = -\langle \vec{AC}, \vec{AB} \rangle + bc \quad | \cdot (-1)$$

$$\Rightarrow -2(p-b)(p-c) = \langle \vec{AC}, \vec{AB} \rangle - bc.$$

$$\boxed{10}. \quad \langle a, b \rangle = \begin{cases} 0, & a=0 \text{ or } b=0 \\ |a| \cdot |b| \cdot \cos \angle(a, b) & \text{otherwise.} \end{cases}$$

$$\langle -, - \rangle : V \times V \rightarrow \mathbb{R} \quad (\text{map.})$$

$$1) \text{ bilinear: } (\forall) a, b \in \mathbb{R}, (\forall) u, v, w \in V?$$

$$\langle av + bw, u \rangle = a \langle v, u \rangle + b \langle w, u \rangle$$

$$\langle v, aw + bu \rangle = a \langle v, w \rangle + b \langle v, u \rangle$$

2) symm. $(\forall) v, w \in V^2$.

$$\langle v, w \rangle = \langle w, v \rangle.$$

3) pos-def. $(\forall) v \in V^2$:

$$v \neq 0 \Rightarrow \langle v, v \rangle \geq 0.$$

4) n -angles, unit-lengths.

$(\forall) v, w \in V^2$:

$$v \perp w \Leftrightarrow \langle v, w \rangle = 0.$$

$$|v| = 1 \Leftrightarrow \langle v, v \rangle = 1$$

Prove bilinearity:

$$\langle a, b \rangle = |a| \cdot |b| \cdot \cos \angle(a, b)$$

$$= |b| \cdot |a| \cdot \frac{\text{pr}_{\perp b} a}{|a|}$$

$$= |b| \cdot \underbrace{\text{pr}_{\perp b} a}_{\text{ct.}} \cdot \underbrace{|a|}_{\text{dim. fct. in } b}.$$

?

Prove symmetry

$$\langle a, b \rangle = |a| \cdot |b| \cdot \cos \angle(a, b).$$

$$\langle b, a \rangle = |b| \cdot |a| \cdot \cos \angle(a, b).$$

multipl. of real nr. is comm.

For def.

$$\langle a, a \rangle > 0$$

$$\langle a, a \rangle = |a| \cdot |a| \cdot \cos 0 = |a|^2 \geq 0.$$

$$\langle a, a \rangle = 0 \Leftrightarrow |a|^2 = 0 \Leftrightarrow |a| = 0 \Leftrightarrow a = 0$$

R-angles, unit lengths.

$$\bullet \langle a, b \rangle = |a| \cdot |b| \cdot \cos \angle(a, b).$$

$$\Leftrightarrow a \perp b \Rightarrow \angle(a, b) = 90^\circ \Rightarrow \cos \angle(a, b) = 0.$$

$$\Rightarrow \langle a, b \rangle = 0.$$

$$\Leftrightarrow \langle a, b \rangle = 0, a, b \neq 0$$

$$\Rightarrow |a| \cdot |b| \cdot \cos \angle(a, b) = 0 \Rightarrow \cos \angle(a, b) = 0$$

$$\Rightarrow \angle(a, b) = 90^\circ.$$

$$\bullet \langle a, a \rangle = |a|^2.$$

$$|a| = 1 \Rightarrow \langle a, a \rangle = 1.$$

$$\Rightarrow |a|^2 = 1.$$

$$\Rightarrow |a| = 1.$$

$$\langle a, a \rangle = 1 \Leftrightarrow |a| = 1.$$

$$! \quad \text{Pr}_{a^\perp} b = \frac{\langle a, b \rangle}{\langle a, a \rangle} a = \text{pr}_{a^\perp} b \cdot \frac{a}{|a|}$$

$$\text{Pr}_{b^\perp} a = \frac{\langle a, b \rangle}{\langle b, b \rangle} b = \text{pr}_{b^\perp} a \cdot \frac{b}{|b|}$$

$$\text{III. } \langle a, b \rangle = \sum_{i=1}^n a_i b_i$$

12. Gram-Schmidt.

$$e'_1 = e_1$$

$$e'_2 = e_2 - \frac{\langle e'_1, e_2 \rangle}{\langle e'_1, e'_1 \rangle} e'_1 = e_2 - \text{Pr}_{e'_1}^\perp e_2$$

$$e'_3 = e_3 - \frac{\langle e'_1, e_3 \rangle}{\langle e'_1, e'_1 \rangle} e'_1 - \frac{\langle e'_2, e_3 \rangle}{\langle e'_2, e'_2 \rangle} e'_2 = e_3 - \text{Pr}_{e'_1}^\perp e_3 - \text{Pr}_{e'_2}^\perp e_3$$

$$\Rightarrow \mathcal{B} = \left\{ \frac{e'_1}{\|e'_1\|}, \dots, \frac{e'_n}{\|e'_n\|} \right\}.$$

$$\mathcal{B}_K = (e_1, \dots, e_K).$$

$$\mathcal{B}_{K'} = (e'_1, \dots, e'_K) \text{ orthonormal.}$$

$$\bullet K=1 \checkmark$$

$$\bullet K \geq 1: e'_{K+1} = e_{K+1} - \underbrace{\sum_{j=1}^K \text{Pr}_{e'_j}(e_{K+1})}_{\text{not } e_{K+1}}$$

• Linear comb:

$$\frac{\langle e'_j, e_{K+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot e'_j$$

• Lin. ind + gen. space $\Rightarrow (e'_1, \dots, e'_K, e'_{K+1})$ basis.

$$\langle e'_{K+1}, e'_i \rangle = \langle e_{K+1} - \sum_{j=1}^K \frac{\langle e'_j, e_{K+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot e'_j, e'_i \rangle$$

$$= \langle e_{K+1}, e'_i \rangle - \sum_{j=1}^K \frac{\langle e'_i, e_{K+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot \underbrace{\langle e'_j, e'_i \rangle}_{\substack{\text{if } i=j, 1 \\ \text{otherwise, } 0}}$$

$$= \langle e_{k+1}, e_i' \rangle - \frac{\langle e_i', e_{k+1} \rangle}{\langle e_i', e_i' \rangle} \cdot \langle e_i', e_i' \rangle = 0$$

③. $\mathcal{H}: a_1 x_1 + \dots + a_n x_n = b.$

Fix $Q \in \mathcal{H} \Rightarrow a_1 q_1 + \dots + a_n q_n = b. \quad (1)$

Fix $P \in \mathcal{H} \Rightarrow a_1 p_1 + \dots + a_n p_n = b \quad (2)$

$(2) - (1) \Rightarrow P - Q = \vec{QP} = a_1 (p_1 - q_1) + \dots + a_n (p_n - q_n) = 0$

$m = (a_1, \dots, a_n) \Rightarrow P \in \mathcal{H} \Leftrightarrow \langle m, \vec{QP} \rangle = 0$

$\Leftrightarrow m \perp \vec{QP}.$

④. dist. from point $\rightarrow$ hyperplane.

$d(P, \mathcal{H}) = \frac{|a_1 p_1 + \dots + a_n p_n + a_{n+1}|}{\sqrt{a_1^2 + \dots + a_n^2}}$

$\mathcal{H}: a_1 x_1 + \dots + a_n x_n + a_{n+1} = 0.$

$Q \in \mathcal{H}: a_1 q_1 + \dots + a_n q_n = -a_{n+1}.$

$m = (a_1, \dots, a_n)$ normal vector.

$d(P, \mathcal{H}) = \left| \text{Proj}_m(\vec{QP}) \right| = \left| \frac{\langle m, \vec{QP} \rangle}{\langle m, m \rangle} m \right| =$

$= \left| \frac{\langle m, \vec{QP} \rangle}{\langle m, m \rangle} \right| \cdot |m| = \left| \frac{\langle m, \vec{QP} \rangle}{|m|^2} \right| \cdot |m| = \left| \frac{\langle m, \vec{QP} \rangle}{|m|} \right| =$

$= \left| \frac{\langle m, \vec{OP} - \vec{OQ} \rangle}{|m|} \right| = \left| \frac{\langle m, \vec{OP} \rangle - \langle m, \vec{OQ} \rangle}{|m|} \right| =$

$= \left| \frac{a_1 p_1 + \dots + a_n p_n - (a_1 q_1 + \dots + a_n q_n)}{|m|} \right|$

$= \frac{|a_1 p_1 + \dots + a_n p_n - (-a_{n+1})|}{\sqrt{a_1^2 + \dots + a_n^2}} = \frac{|a_1 p_1 + \dots + a_{n+1}|}{\sqrt{a_1^2 + \dots + a_n^2}} \quad (14)$

7. Points in $\mathbb{E}^2$, equidistant from given affine subspace.

• case 1: affine subspace = point. (dim = 0)

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, Q) = \alpha\} = \\ &= \{P(x, y) \mid \sqrt{(x - x_Q)^2 + (y - y_Q)^2} = \alpha\} = \text{circle}. \end{aligned}$$

• case 2: affine subspace = line

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, \ell) = \alpha\} \\ &= \{P(x, y) \mid \frac{|ax + by + c|}{\sqrt{a^2 + b^2}} = \alpha\} \\ &= \{P \mid ax + by + c = \pm \alpha \cdot \sqrt{a^2 + b^2}\} \\ &= \{P \mid ax + by + c + \alpha \sqrt{a^2 + b^2} = 0\} \cup \\ &\quad \cup \{P \mid ax + by + c - \alpha \sqrt{a^2 + b^2} = 0\} \\ &= 2 \text{ lines} \Rightarrow \parallel. \end{aligned}$$

16. $\gamma(v)$ dim 2.

$$\gamma(v) = (a, b)$$

$$v(x, y) \neq 0 \Rightarrow x^2 + y^2 \neq 0. \quad (1) \quad \begin{aligned} &\Rightarrow by = -ax \\ &y = -\frac{ax}{b} \\ &y^2 = -\frac{a^2 x^2}{b^2} \end{aligned}$$

$$\begin{cases} ax + by = 0 \quad (1) & \Rightarrow b = -\frac{ax}{y}, y \neq 0. \\ x^2 + y^2 = a^2 + b^2 \quad (\text{same length}) \end{cases}$$

$$\Rightarrow x^2 + y^2 = a^2 + a^2 \cdot \frac{x^2}{y^2} = a^2 \left(1 + \frac{x^2}{y^2} \right) =$$

$$= a^2 \cdot \left(\frac{x^2 + y^2}{y^2} \right)$$

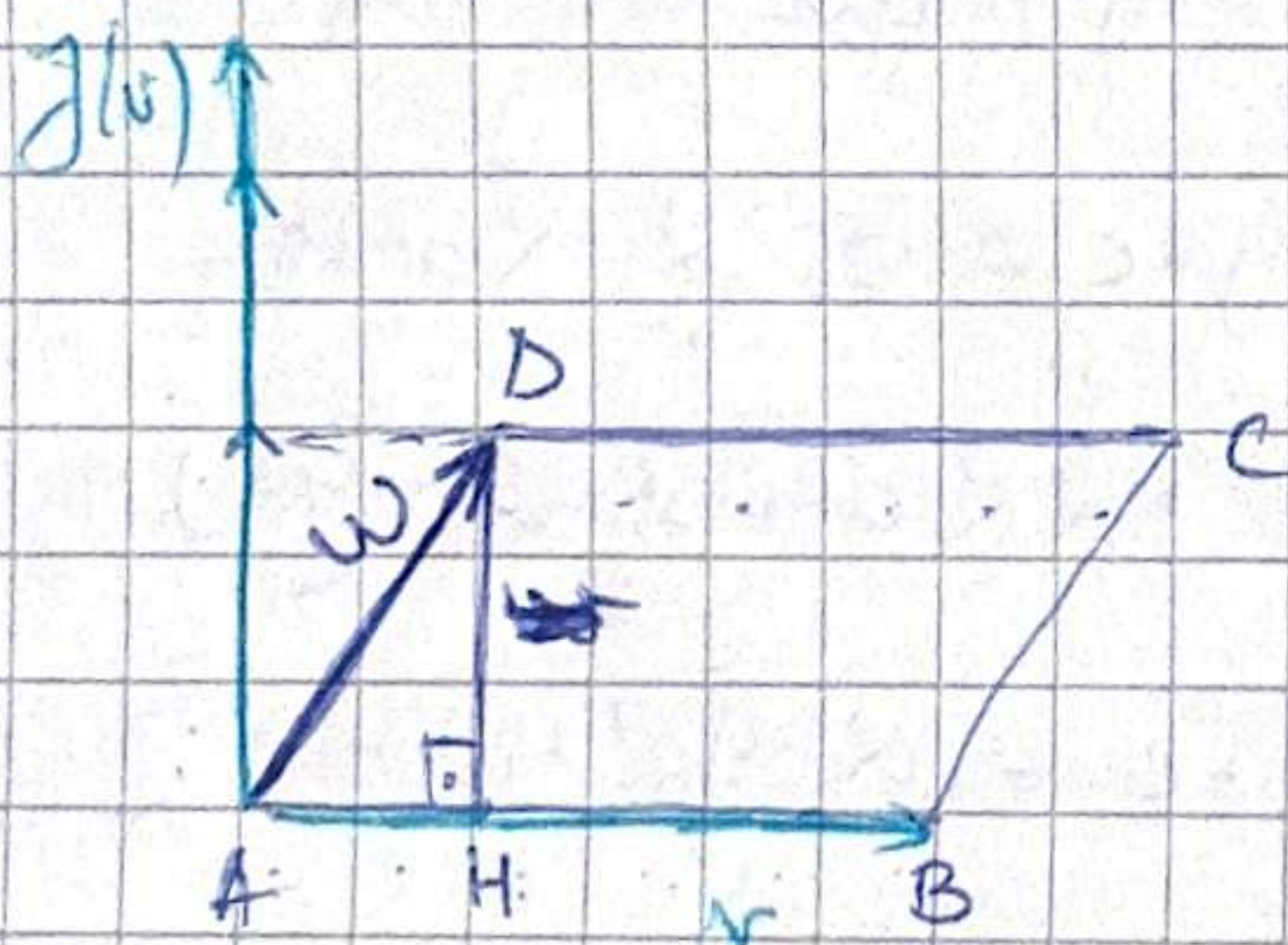
$$= a^2 \cdot \frac{x^2 + y^2}{\frac{a^2 x^2}{b^2}} = a^2 - \frac{x^2 + y^2}{b^2} \Rightarrow a^2 = y^2 \Rightarrow a = \pm y \quad (15)$$

$$\Rightarrow \begin{vmatrix} x & a \\ y & b \end{vmatrix} = \begin{vmatrix} x & \pm y \\ y & \mp y - \frac{x}{y} \end{vmatrix} = \dots$$

$\downarrow \quad \downarrow$
 $v \quad j(w)$

17. A, B, D of ABCD parallelogram.

$$A_{ABCD} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix} = \begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix}$$



$$v(v_x, v_y) = \vec{AB}, \quad j(w)(-v_y, v_x).$$

$$w(w_x, w_y) = \vec{AD}.$$

$$\begin{aligned} A_{ABCD} &= b \cdot h = |\vec{AB}| \cdot |\vec{AB}^\perp| = |v| \cdot |\text{Pr}_{j(w)} w| = \\ &= |v| \cdot \left| \frac{\langle j(w), w \rangle}{\langle j(w), j(w) \rangle} \right| = |v| \cdot \left| \frac{\langle j(w), w \rangle}{|j(w)|} \right| = \\ &= |\langle j(w), w \rangle| = | \langle (-v_y, v_x), (w_x, w_y) \rangle | = \\ &= | -v_y w_x + v_x w_y | = \begin{vmatrix} v_x & w_y \\ v_y & w_x \end{vmatrix} \end{aligned}$$

$$= \begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix} = \begin{vmatrix} x_B - x_A & y_B - y_A \\ x_D - x_A & y_D - y_A \end{vmatrix}^2$$

$$v^2 a^2 + \dots + a^2 n^2$$

(16)

$$\begin{vmatrix} x_A & y_A & 1 \\ x_B - x_A & y_B - y_A & 0 \\ x_D - x_A & y_D - y_A & 0 \end{vmatrix} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix}$$

18. $\triangle ABC$;

? $u \times v = v \times w = w \times u = A$

? Sim law.

$$u + v + w = 0 \Rightarrow w = -(u + v)$$

$$v \times w = v \times (-(u + v)) = -v \times u - \overbrace{v \times v}^0 = \underline{u \times v}$$

$$w \times u = -(u + v) \times u = -\underbrace{u \times u}_0 - v \times u = \underline{u \times v}$$

$$(1) \quad |u \times v| = |u| \cdot |v| \cdot \sin(\widehat{u, v}) = A = |v \times w| = |w \times u|$$

$$(2) \quad |v \times w| = |v| \cdot |w| \cdot \sin(\widehat{v, w}) = A$$

$$(3) \quad |w \times u| = |w| \cdot |u| \cdot \sin(\widehat{w, u}) = A$$

$$(1) = (2) \Rightarrow | : |v|$$

$$\Rightarrow |u| \cdot \sin \beta = |w| \cdot \sin \alpha$$

$$\Rightarrow \frac{|u|}{\sin \alpha} = \frac{|w|}{\sin \beta}$$

$$\Rightarrow T \cdot \sin \cdot \checkmark$$

$$(2) = (3) \Rightarrow | : |w|$$

$$\Rightarrow \frac{|v|}{\sin \beta} = \frac{|u|}{\sin \alpha}$$

[19]

since $x \rightarrow$ box pt.cosine $\rightarrow$ scalar pt.

$$(\sin \angle_{or}(v, w))^2 + (\cos \angle_{or}(v, w))^2 = 1.$$

$$\sin \angle_{or}(v, w) = \frac{[v, w]}{|v| \cdot |w|} = \frac{2 \cdot \text{Area}(ABC)}{|AB| \cdot |AC|}$$

$$\cos \angle_{or}(v, w) = \frac{\langle v, w \rangle}{|v| \cdot |w|}$$

$$(+)$$

$$= \frac{[v, w]^2 + \langle v, w \rangle^2}{|v|^2 \cdot |w|^2}$$

$$= \frac{(v_x w_y - v_y w_x)^2 + (v_x w_x + v_y w_y)^2}{|v|^2 \cdot |w|^2}$$

$$= \frac{(v_x^2 + v_y^2) \cdot (w_x^2 + w_y^2)}{|v|^2 \cdot |w|^2} = \frac{|v|^2 \cdot |w|^2}{|v|^2 \cdot |w|^2} = 1.$$

$$[v, w] = \begin{vmatrix} v_x & w_x \\ v_y & w_y \end{vmatrix} = \begin{vmatrix} v_x & w_y \\ w_x & w_y \end{vmatrix} = v_x w_y - v_y w_x.$$

$$\langle v, w \rangle = \langle (v_x, v_y), (w_x, w_y) \rangle = v_x w_x + v_y w_y.$$

[20]

$$u = (u_1, u_2, u_3)$$

$$v = (v_1, v_2, v_3)$$

$$u \times v = (u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1)$$

$$v \times u = (v_2 u_3 - v_3 u_2, v_3 u_1 - v_1 u_3, v_1 u_2 - v_2 u_1)$$

$$= -(u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, \dots)$$

$$|w_x w_y|$$

$$|AD + AH|$$

$$|v \cdot w|$$

(18)

6

[21.] Jacobi. $(\forall) a, b, c \in V^3$

$$(a \times b) \times c + (b \times c) \times a + (c \times a) \times b = 0;$$

$$(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x \quad \text{Double cross formula}$$

$$(a \times b) \times c = \langle a, c \rangle b - \langle b, c \rangle a$$

$$(b \times c) \times a = \langle b, a \rangle c - \langle c, a \rangle b$$

$$(c \times a) \times b = \langle c, b \rangle a - \langle a, b \rangle c$$

(+)

$$= 0.$$

[22.] Lagrange.

$$\langle a \times b, c \times d \rangle = \langle a, c \rangle \cdot \langle b, d \rangle - \langle b, c \rangle \cdot \langle a, d \rangle$$

$$= \begin{vmatrix} \langle a, c \rangle & \langle a, d \rangle \\ \langle b, c \rangle & \langle b, d \rangle \end{vmatrix}$$

$$(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x \quad \text{Double cross formula.}$$

$$x = a \times b$$

$$x = c \quad \Rightarrow (c \times d) \times (a \times b) =$$

$$y = d \quad = \cancel{c \times d} = \langle c, a \times b \rangle d - \langle d, a \times b \rangle c$$

$$z = a \times b \quad =$$

$$(c \times d) \times (a \times b) \perp (a \times b)$$

$$\Rightarrow (a \times b) \times d = \langle a, d \rangle b - \langle b, d \rangle a.$$

$$\langle a \times b, c \times d \rangle = \langle (a \times b) \times d, c \rangle =$$

✓

$$? = \langle \langle a, d \rangle b - \langle b, d \rangle a, c \rangle$$

$$\boxed{23} \cdot (a \times b) \times (c \times d) = b[a, c, d] - a[b, c, d] = \\ = c[a, b, d] - d[a, b, c]$$

$$[a, b, c] = \langle a \times b, c \rangle$$

$$(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x$$

$$\langle c, a \times b \rangle = [c, a, b] = [a, b, c]$$

$$\langle d, a \times b \rangle = [d, a, b] = [a, b, d]$$

$$\left. \begin{array}{l} x = a \\ y = b \\ z = c \times d \end{array} \right\} \Rightarrow \begin{aligned} (a \times b) \times (c \times d) &= \langle a, c \times d \rangle b - \\ &\quad - \langle b, c \times d \rangle a = [a, c, d]b - [b, c, d]a \\ &= b[a, c, d] - a[b, c, d] \end{aligned}$$

$$\begin{aligned} ? \Rightarrow (c \times d) \times (a \times b) &= -\langle c, a \times b \rangle d + \\ &\quad + \langle d, a \times b \rangle c = -[c, a, b]d + \\ &\quad + [d, a, b]c \\ &= -d[c, a, b] + c[d, a, b] \\ &= c[a, b, d] - d[a, b, c] \end{aligned}$$

$\boxed{24}$. dist. from point to line in $\mathbb{E}^3$.

dist. between two parallel lines in $\mathbb{E}^2$.

(2)

(20)

$$A_{\text{par}} = b \cdot h = |v| \cdot d(P, Q) = |\vec{AP} \times v|$$

$$\Rightarrow d(P, Q) = \frac{|\vec{AP} \times v|}{|v|}$$

$$A(x_A, y_A, z_A), P(x_P, y_P, z_P), v(v_x, v_y, v_z).$$

$$|\vec{AP} \times v| = \begin{vmatrix} i & j & k \\ x_P - x_A & y_P - y_A & z_P - z_A \\ v_x & v_y & v_z \end{vmatrix}$$

$$d(P, Q) = \frac{\begin{vmatrix} i & j & k \\ x_P - x_A & y_P - y_A & z_P - z_A \\ v_x & v_y & v_z \end{vmatrix}}{\sqrt{v_x^2 + v_y^2 + v_z^2}}$$

$$l \parallel l'; B \in l'$$

$$\Rightarrow d(l, l') = d(B, l) = \frac{|\vec{AB} \times v|}{|v|}$$

25. common line of two skew lines ($\mathbb{E}^3$).

$$l: A + t \cdot v$$

$$l': B + s \cdot w$$

$$l \nparallel l' \Rightarrow \text{skew} \Rightarrow a = v \times w \neq \underline{0}.$$

Take plane $\bar{u}$ through A, $\parallel v$ and a :

$$l \subset \bar{u} \Rightarrow m = \underbrace{(v \times w)}_{=a} \times v = a \times v.$$

Take plane $\bar{u}'$ through B, $\parallel w$ and a :

$$l' \subset \bar{u}' \Rightarrow m' = \underbrace{(v \times w)}_{=a} \times w = a \times w.$$

$$d = \bar{u} \cap \bar{u}'$$

$n \times m' = (a \times v) \times (a \times w)$ proportional to

$a = v \times w \Rightarrow a = \text{dir. vector of } d.$

$$a \perp v \Rightarrow d \perp l$$

$$a \perp w \Rightarrow d \perp l'$$

$$d = \bar{u} \cap \bar{u}'$$

$$\Rightarrow d = l \cap l' ?$$

$$\bar{u}: \begin{vmatrix} x-x_A & y-y_A & z-z_A \\ v_x & v_y & v_z \\ a_x & a_y & a_z \end{vmatrix} = 0$$

$$\bar{u}': \begin{vmatrix} x-x_B & y-y_B & z-z_B \\ w_x & w_y & w_z \\ a_x & a_y & a_z \end{vmatrix} = 0$$

[26] Box prod $\Leftrightarrow$ cross prod, scalar prod.

$V_{\text{parallelepiped}} = \text{abs value of } \dots$

$[a, b, c] = \langle a \times b, c \rangle \rightarrow \text{scalar triple prod.}$

$|a \times b| = \text{area of parallelogram gen. by } a, b.$

$\theta \rightarrow \text{angle between } a \times b \text{ and } c.$

$$\langle a \times b, c \rangle = |a \times b| \cdot \underbrace{|c| \cdot \cos \theta}_{\text{height}}$$

$$\begin{aligned}
 V &= A \times h = \underbrace{|a \times b|}_b \cdot \underbrace{|c|}_h \cdot \cos \theta \\
 &= \langle a \times b, c \rangle \\
 &= [a, b, c].
 \end{aligned}$$

$$\begin{aligned}
 \bullet (a, b, c) \text{ r-or} &\Rightarrow + \\
 &\quad \text{l-or} \Rightarrow -
 \end{aligned}$$

$$\Rightarrow V = |[a, b, c]| = |\langle a \times b, c \rangle|.$$

27. Dist. between two skew lines in $\mathbb{E}^3$.

$$\begin{cases}
 l: A + t \cdot v \\
 l': B + s \cdot w
 \end{cases}$$

$v \nparallel w$ (skew)

$$\bullet m = v \times w$$

$$\bullet \vec{AB} = B - A.$$

$$d(l, l') = \frac{|\langle \vec{AB}, v \times w \rangle|}{|v \times w|} = \frac{|\langle \vec{AB}, m \rangle|}{|m|} =$$

$$= \frac{|[AB, v, w]|}{|v \times w|} = \frac{\text{BOX}}{\text{CROSS}}.$$